

Computational Model of Secondary Palate Fusion and Disruption

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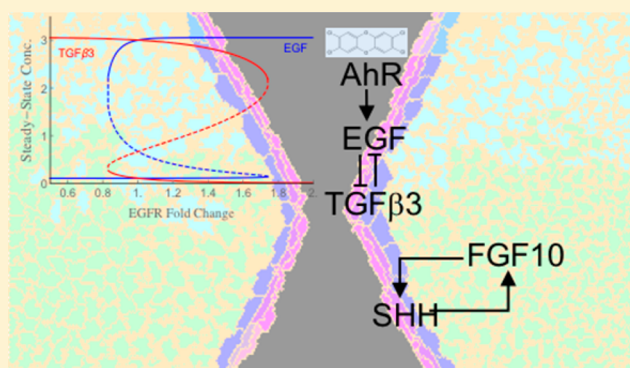
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Supporting Information

ABSTRACT: Morphogenetic events are driven by cell-generated physical forces and complex cellular dynamics. To improve our capacity to predict developmental effects from chemical-induced cellular alterations, we built a multicellular agent-based model in CompuCell3D that recapitulates the cellular networks and collective cell behavior underlying growth and fusion of the mammalian secondary palate. The model incorporated multiple signaling pathways (TGF β , BMP, FGF, EGF, and SHH) in a biological framework to recapitulate morphogenetic events from palatal outgrowth through midline fusion. It effectively simulated higher-level phenotypes (e.g., midline contact, medial edge seam (MES) breakdown, mesenchymal confluence, and fusion defects) in response to genetic or environmental perturbations. Perturbation analysis of various control features revealed model functionality with respect to cell signaling systems and feedback loops for growth and fusion, diverse individual cell behaviors and collective cellular behavior leading to physical contact and midline fusion, and quantitative analysis of the TGF/EGF switch that controls MES breakdown—a key event in morphogenetic fusion. The virtual palate model was then executed with theoretical chemical perturbation scenarios to simulate switch behavior leading to a disruption of fusion following chronic (e.g., dioxin) and acute (e.g., retinoic acid) chemical exposures. This computer model adds to similar systems models toward an integrative “virtual embryo” for simulation and quantitative prediction of adverse developmental outcomes following genetic perturbation and/or environmental disruption.



1. INTRODUCTION

Understanding how tissues and organs are shaped during embryogenesis is a central question in developmental biology. Morphogenetic events are driven by cell-generated physical forces and complex cellular dynamics. The key drivers are individual cell behaviors, mitosis, migration, differentiation, adhesion, shape-change, apoptosis, and extracellular matrix (ECM) remodeling, but the realized series of morphogenetic events depends on cells' coordinated and collective behavior.^{1,2} Individual cells may display incorrect or inappropriate behavior, either spontaneously or as a result of genetic and/or environmental factors, and yet gene regulatory networks exist that buffer embryos against microshifts in cell fate or specification.³ Whether such canalization ensures normal development will depend on the degree and nature of perturbation and on the resiliency of the affected systems: teratogenesis occurs when embryos encounter unexpected environmental stressors that exceed the system's buffering limits.⁴ As such, our capacity to predict developmental

effects from cellular alterations is limited by extant knowledge of the dynamic control and resiliency of cellular networks.

Here, we describe a multicellular agent-based model that recapitulates the cellular networks and collective cell behavior underlying a particular morphogenetic event, growth and fusion of the mammalian secondary palate, and use this computational (*in silico*) model to further mechanistic understanding and predictive modeling of developmental toxicity leading to cleft palate. The mammalian secondary palate develops from two separate bilaterally paired shelves that meet and fuse into a continuous anatomical structure separating the oral and nasal cavities.⁵ Disruption of these events by genetic and/or environmental factors can lead to cleft palate, an important human birth defect that affects 5 to 20 in 10,000 live births.^{1,6} In

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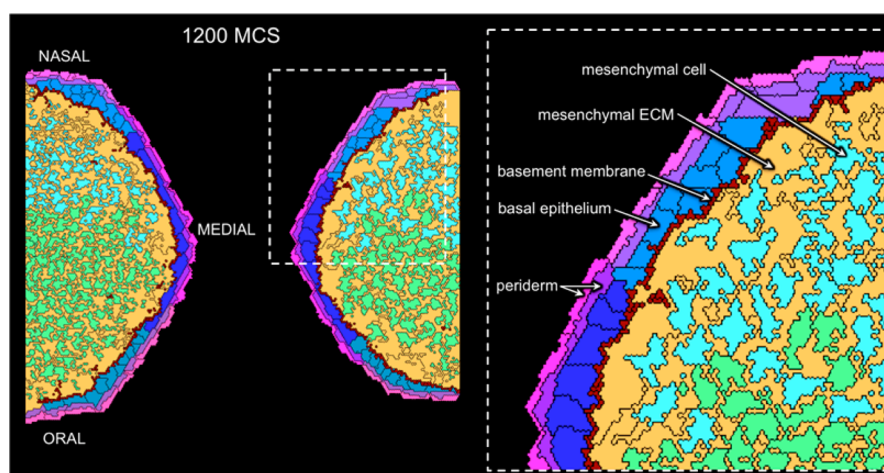


Figure 1. Layout of the virtual palate model (prefusion, E13–E14) in a coronal plane. Mesenchymal cells (green and cyan) are randomly embedded in a hyaluronate-rich ECM (orange). The color-coding for mesenchymal cells denotes their origin from either the oral (green) or nasal (cyan) half of each palate shelf. The overlying ectoderm is represented as a columnar epithelium (shades of blue) covered by squamous periderm cells that are polarized with separate apical and basal compartments (shades of pink and magenta, respectively). Ectoderm cells with darker shading are those covering the medial edge. A basement membrane (dark red) lies along the ectoderm–mesenchyme border, and the palatal shelves are surrounded by a fluid filling the oral–nasal cavity (black). Cell types are labeled in the expanded inset.

animal studies and human epidemiology, cleft palate is observed following prenatal exposure to many compounds, including glucocorticoids,^{7–9} valproic acid,^{10,11} alcohol,^{12,13} and methylmercury.¹⁴ The window of vulnerability for cleft palate is 10–14 days of gestation in the mouse,¹⁵ corresponding to 5–7 weeks gestation in humans.¹⁶ In mammals, as informed by mouse studies, the morphogenetic progression spans from initial budding of palatal shelves from the maxillary processes (E12.5) to outgrowth vertically downward alongside the tongue, reorientation to a horizontal position above the tongue, contact of the shelves' medial edge epithelium (MEE) to form a midline epithelial seam (MES), and dissolution of the MES to allow fusion of right–left palate rudiments (E15.5).¹⁷ Multiple signaling pathways are involved, including those for transforming growth factor beta (TGF β), bone morphogenetic proteins (BMPs), fibroblast growth factors (FGFs), and sonic hedgehog (SHH).¹⁸ These pathways mediate epithelial–mesenchymal interactions to drive outgrowth and fusion of the palatal processes.^{17–19}

In an alternate emerging paradigm, computational toxicology models make predictions for apical end points such as cleft palate based on vast amounts of cellular and molecular effects data now on hand from profiling thousands of chemicals across hundreds of high-throughput and high-content screening assays, including human stem cells and embryonic zebrafish.²⁰ A significant challenge is building models like the one presented here that use biological frameworks to recapitulate complex signaling networks in the embryo. Although critical aspects of *in vivo* palate development may await discovery, computer simulation with an agent-based model can be used to reconstruct a morphogenetic series of events from the bottom-up, cell-by-cell, and interaction-by-interaction. The capacity to simulate a self-organizing biological system *in silico* offers a heuristic solution to predict tissue-level effects of environmental exposure(s) in a model that approximates the extant embryology and accommodates future discoveries. The model presented here for palatal development (cleft palate) adds to previous models for angiogenesis (angiodyplasia)²¹ and urethral fusion (hypospadias)²² toward an array of systems comprising a “virtual embryo” for simulation

and quantitative prediction of adverse developmental outcomes following genetic perturbation and/or environmental disruption.

2. METHODS AND IMPLEMENTATION

2.1. Model Scope. The virtual palate model was constructed using CompuCell3D, version 3.7.4 (CC3D; <http://CompuCell3D.org>) with scripting in Python (v 2.7). CompuCell3D is an open-source environment for simulating the spatiotemporal dynamics of multicellular systems given specified cell behaviors, regulatory signals, and physical properties. The model begins with an initial tissue configuration corresponding to the onset of palate shelf formation and recapitulates growth and fusion of the mammalian secondary palatal processes through midline MEE contact and MES dissolution. Specifications were developed from relevant information in the extant literature, focusing on several key signaling pathways (TGF β , SHH, FGF, EGF, and BMP), and their regulation of differentiation and cell behaviors, e.g., proliferation, apoptosis, polarization, ECM secretion, and epithelial-to-mesenchymal transitions.

An appropriately abstracted model and boundary conditions were defined for gestational days E12.5 to E16.0 in the mouse to recapitulate the sequence of events spanning initial budding of palatal shelves from the maxillary processes (E12.5) to fusion of right–left palate rudiments (E15.5), timed with regard to conventional nomenclature of E0 being the onset of gestation.¹⁷ Anatomically, this progression includes mesenchymal proliferation and ECM accumulation (E12.5–13.5); shelf reorientation to a horizontal position above the tongue (E13.5–14.5); contact of the rudiments at their medial edges to form a midline epithelial seam (MES); and MES breakdown to allow right-to-left confluence of mesenchyme (E14–15.5). We modeled these stages using an idealized two-dimensional (2D) cross-section that includes right and left palatal process growing medially and surrounded by a fluid medium reflecting the primitive oral–nasal cavity (Figure 1). The model explicitly represents a coronal cross-section through the anterior region of the secondary palatal processes. It recapitulates all of the events described above except reorientation of the shelves from vertical to horizontal. Palatal shelves in the computer model grow horizontally from the start, which reflects tissue growth kinetics to bring the right and left shelves into contact, but is an abstraction of the reoriented condition that requires a third dimension and is left for future implementation.

CompuCell3D is based on a cellular Potts model that treats individual cells as autonomous agents interacting in a shared physical environment, modeled as a discrete lattice, in which each cell occupies multiple lattice sites.^{23,24} The initial configuration of the virtual palate model consists of

Table 1. Matrix of Contact Energies in the Virtual Palate Model^a

"cell" type	MESENC	ECM	BM (int)	EPITH	PERI_B	PERI_A (int)	EXT
MESENC	3	−1	6	16	30	30	100
ECM		0	6	16	30	30	100
BM (int)			8 (6)	6	20	30	100
EPITH				4	10	30	100
PERI_B (int)					10	30 (2)	30
PERI_A						10	6
EXT							0

^aPrimary entries are for J_{ext} with energies for the two possible intracellular contacts in J_{int} listed in parentheses. Abbreviations: MESENC, mesenchyme; ECM, extracellular matrix; BM, basement membrane; EPITH, epithelium; PERI_B, periderm basal compartment; PERI_A, periderm apical compartment; and EXT, surrounding external fluid.

two circular segments (radius = 46.5 lattice sites or pixels; segmenting chord positioned 19 pixels off center; 1 pixel is approximately 1 μm) that protrude from the vertical borders of a 200×400 hexagonal lattice. Each semicircular segment represents the initial budding of a palatal shelf from the left or right side of the embryo with an ectodermal epithelium of 20 epithelial cells and 10 periderm cells encasing an interior 35-cell, ECM-free mesenchyme. This initial cross-section of the midanterior palate is roughly quarter-scaled to the biology: cross sections of the mouse midanterior palate have about 150 cells and very little extracellular space at E12.5.²⁵

Model dynamics follow a Metropolis Monte Carlo algorithm that repeatedly attempts to change which cell occupies randomly chosen lattice sites.²⁶ The success of attempted changes is governed by corresponding changes in a pseudoenergy function (ΔE ; see below for details): favorable pseudoenergy changes ($\Delta E \leq 0$) are always successful; unfavorable changes ($\Delta E > 0$) are successful with probability $P = e^{-\Delta E/T_{\text{eff}}}$ where T_{eff} is the fluctuation amplitude or effective temperature. This cell-specific value reflects motility of that particular cell type and adds a stochastic element to the model framework. Dynamics were discretized in Monte Carlo Steps (MCS), each MCS corresponding to $200 \times 400 = 80,000$ attempted changes. Each simulation began with a 200-MCS equilibration period and then continued to 6000 MCS.

2.2. Cell and ECM Types. The model includes three primary cell types, mesenchymal, epithelial, and periderm, each with multiple subtypes (refer to Figure 1 for color schema). Mesenchymal cells can be of oral, nasal, or EMT subtype based on the spatial location of the cell's earliest predecessor; cells of EMT subtype are those that began as epithelial but then underwent an epithelial-to-mesenchymal transition. Epithelial and periderm cells can similarly be of oral, medial, or nasal subtype. In addition, periderm cells are polarized into two subcellular compartments, apical and basal, that the model treats as paired agents.

The model also includes two types of extracellular matrix, mesenchymal ECM and basement membrane (BM), that are treated as generalized "cells" in CC3D. Elements of these types are assigned pseudoenergy terms that govern their interactions with cells (see below), but they are not granted the same agency as cells.

2.3. Pseudoenergy Function. The pseudoenergy function governing temporal evolution of the model has terms for interfacial contact energies ($E_{\text{interface}}$), shape constraints (E_{shape}), and focal point plasticity ($E_{\text{focal-point plasticity}}$). The paragraphs below describe each of these terms and detail which pseudoenergy parameters are fixed and which vary dynamically as the model evolves.

The contact energy terms determine how cells, ECM, and BM segregate. The pseudoenergy of any generalized cell–cell interface is proportional to its length with a proportionality constant that depends on the two contacting cell types.²⁶ The sum over all interfaces can be written as

$$E_{\text{interface}} = \sum_{\text{neighbor sites } i, j} (1 - \delta_{\rho_i, \rho_j}) [\delta_{\sigma_i, \sigma_j} J_{\text{int}, \tau_i, \tau_j} + (1 - \delta_{\sigma_i, \sigma_j}) J_{\text{ext}, \tau_i, \tau_j}] \quad (1)$$

where (ρ_i, ρ_j) , (σ_i, σ_j) , and (τ_i, τ_j) are respectively the generalized cell IDs, cluster IDs, and cell types at neighboring lattice sites i and j ; δ is the

Kronecker delta function; and J_{int} and J_{ext} are matrices of cell-type-dependent interfacial energies (listed in Table 1). J_{ext} is applicable to contact between different cells or between a cell and the ECM, basement membrane, or surrounding medium. J_{int} is applicable only for contact between subcellular compartments of a single generalized cell, e.g., the apical and basal compartments of a periderm cell, which have the same cluster ID. The primary entries in Table 1 are for J_{ext} , but it also includes entries for the model's only two possible intracellular contacts: $J_{\text{int}, \text{Peri}_A\text{-Peri}_B}$ and $J_{\text{int}, \text{BM-BM}}$. These energies correspond biologically to a combination of cortical tension and surface adhesion between a given pair of generalized cell types: smaller energies in Table 1 denote more favorable types of contacts. The two contact energy matrices are defined in the code at the start of a simulation and remain fixed throughout.

The shape constraint terms influence cell volume and shape through quadratic terms that penalize deviations from morphological target values:^{24,27}

$$E_{\text{shapes}} = \sum_{\text{generalized cells } k} \lambda_{\text{vol}, k} \times (V_k - V_{\text{target}, k})^2 + \lambda_{\text{surface}, k} \times (S_k - S_{\text{target}, k})^2 + \lambda_{\text{length}, k} \times (L_k - L_{\text{target}, k})^2 \quad (2)$$

where V , S , and L are, respectively, the cell volume (area in 2D), surface area (perimeter in 2D), and major axis length. Each shape measure has a constant associated stiffness (λ_{vol} , λ_{surface} , or λ_{length}) and a dynamic cell-specific target value. If a particular constraint is not used for a given cell, then that constraint's stiffness is zero. When cell behaviors in a simulation change a cell's target volume, the model automatically updates that cell's target surface area and target length (if applicable) to new values that depend on the new target volume. For periderm cells, the updating relationships are set to favor elongated shapes: $S_{\text{target}} = 0.7V_{\text{target}}$ and $L_{\text{target}} = 0.3V_{\text{target}}$ for unpaired basal compartments; and $S_{\text{target}} = 0.9V_{\text{target}}$ and $L_{\text{target}} = 0.5V_{\text{target}}$ for paired apical and basal compartments. For mesenchymal and epithelial cells, the target surface area is updated to $S_{\text{target}} = 4V_{\text{target}}^{1/2}$ (and no length constraint is used), a relationship that favors cuboidal shapes. This general shape can be seen in the model's epithelial cells, but its mesenchymal cells purposely deviate from cuboidal and are instead setup to favor highly ramified shapes. This is accomplished by setting the MESENC-ECM contact energy negative, which makes the pseudoenergy lower (more favorable) when MESENC cells adopt shapes that increase their contact length with the surrounding ECM. The generalized cells representing ECM and BM have target volumes but not target surface areas or lengths.

The focal point plasticity terms were only used for BM elements to represent the structural elasticity of BM and help maintain it as a thin chain of elements.^{24,27} A focal point plasticity link between the centroids of two BM elements acts as a breakable elastic spring. The energy term for all of these links is given by

$$E_{\text{focal-point plasticity}} = \sum_{\text{linked cells } m, n} \begin{cases} \frac{1}{2} \lambda_{\text{link}} r_{m, n}^2 & \text{for } r_{m, n} \leq r_{\text{break}} \\ 0 & \text{for } r_{m, n} > r_{\text{break}} \end{cases} \quad (3)$$

where λ_{link} is an elastic spring constant, and $r_{m,n}$ is the centroid-to-centroid distance between linked elements m and n . Plasticity arises because the links form probabilistically and break when stretched to $r_{m,n} > r_{\text{break}}$. The spring constant λ_{link} is a global constant in our model, but the linked pairs are dynamic.

2.4. Diffusible Signals. Morphogenesis in the virtual palate model, including palatal outgrowth and midline fusion, is driven by a biological network minimized to specific diffusible signals, juxtacrine signaling via cell–cell contacts, and regulated cell behaviors (Figure 2). On the basis

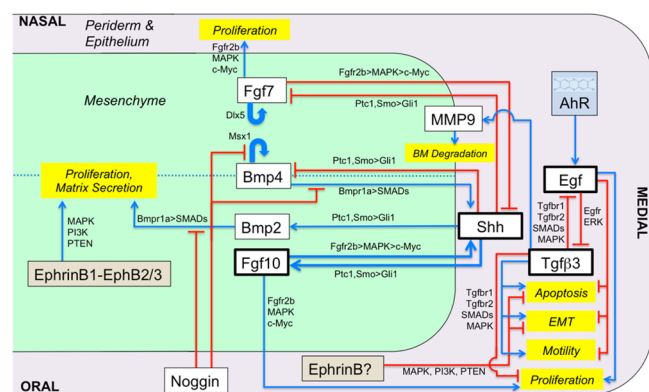


Figure 2. Signaling network driving the virtual palate model. The oral side, nasal side, and medial edge are indicated (not drawn to scale). A minimal signaling network that coordinates several primary pathways (SHH, FGF, BMP, TGF β 3, EGF, Noggin, and EphB) was derived from extant literature. Boxes represent signaling molecules (white) and linked behaviors (yellow) that are explicitly included in the model. Intermediate molecules in the relevant pathways are noted along the connecting arrows.

of a general profile derived from extant literature,^{1,17,28–38} the model represents several key regulatory nodes explicitly. In particular, Sonic Hedgehog (SHH) and FGF10 form a key positive feedback loop for early palatal outgrowth: SHH signals from epithelium to mesenchyme to promote mesenchymal cell proliferation, and FGF10 signals go the other way (mesenchyme to epithelium) to maintain *Shh* expression.^{29,30} Activity of this central loop is modulated by Ephrin juxtacrine signaling³⁴ by FGF7 (on the nasal side),³⁹ and by the interactions among BMP2, BMP4, and Noggin.⁴⁰ The other key signaling module is a mutually inhibitory EGF–TGF β 3 switch. EGF and TGF β 3 signals suppress one another's expression and have opposite effects on ectodermal cell behaviors.^{41–44} In combination with additional Ephrin signals,^{37,45} this switch governs whether cells in the MEE proliferate or die. The behavior of this switch can be modulated by exogenous compounds through cross-talk with pathways mediated by retinoid receptor, glucocorticoid receptor, or aryl hydrocarbon receptor (AhR) systems.^{43,46–56} In addition to these explicitly modeled signals, Figure 2 shows the implicit roles of additional components in each signaling pathway, e.g., the transmission of SHH signaling through patched (Ptc1) and smoothened (Smo),⁵⁷ of FGF signaling through receptors (FGFR2b) in the epithelium,²⁹ and of BMP/TGF β signaling through SMADs.^{17,58} These are not explicitly included in the model but are highlighted to show how their mutation or inhibition would affect the model.

Diffusion and decay parameters for the diffusible signals have not been measured in vivo, so we chose similar values for each molecule with a few notable exceptions. The global diffusion constant for signals regulating mesenchymal growth was 0.5 pixels²/MCS, but that for the ectodermal signals TGF β 3, and EGF was reduced to 20% of this value to reflect local signaling within the epithelial compartment. In addition, cell-type-specific diffusion constants were designated for FGF10, FGF7, and BMP4 to account for their sequestration by the extracellular matrix,^{59,60} slowing their diffusion constant in ECM to 0.2 pixels²/MCS. Finally, the cell-type-specific diffusion constant for SHH was set to zero in the surrounding fluid and ECM to represent how palmitoylation and

cholesterol esterification of SHH render the mature ligand membrane bound, thus requiring cells to create SHH gradients through reiterated uptake and secretion.⁶¹ The rate constant for the decay of each signal was set to 0.005 MCS^{−1}, except for that of FGF7, which required a reduced value of 0.001 MCS^{−1} for its spatial patterning in the simulations to match experimental observations.²⁹

Most regulatory effects on cell behavior were modeled as sigmoidal functions of the diffusible signals. Signal concentrations are thus reported as multiples of each signal's half-maximal activity concentration (AC_{50}). The only receptor levels explicitly included in the model are those for EGF and TGF β 3 when more closely investigating how their mutual inhibition yields a switch-like biochemical circuit. Most simulations used EGFR and TGF β R levels of 1.86 and 2.14, respectively (levels relative to the AC_{50} for EGF and TGF β 3 effects), but a few noted exceptions used levels of 2.86 for both, which led to a more hysteretic and widely bistable switch. The strength of cellular responses to EGF and TGF β 3 signals were then calculated using standard sigmoidal functions of ligand-bound receptor levels (using Hill coefficients of 1 or 4 as detailed for each behavior below). The model did use a ligand–receptor binding affinity (K_D) that was lower than the half-maximal activity concentration of the bound receptor, i.e., $K_D = 0.71 AC_{50}$.

2.5. Model Cell Behaviors and Their Regulation. We used CompuCell3D steppable functions to implement the regulation of eight distinct cell behaviors: cell growth and proliferation, apoptosis, cell differentiation, periderm polarization, cell motility, ECM secretion, BM maintenance, and signal secretion.²⁷ Table 2 lists their assignment to the various cell types and associated effector molecules and a synopsis of their implementation. Although the Python code specified the range of cell fate/state behaviors possible for each cell/agent type, their realization in a simulation was stochastic, that is, locally enabled by probabilities determined by an individual cell's microenvironment.²⁷

2.5.1. Cell Growth and Proliferation. The model implements cell growth and proliferation through semistochastic increases in cells' target volumes; cell divisions then occur when cell volume exceeds a cell-type-specific mitosis trigger volume. All biochemical regulation of proliferation is implemented through control of cell growth. The specific amount added to each cell's target volume is determined every 10th MCS by the cell's regulated growth rate (R_{growth}) that is not stochastic and growth probability (P_{growth}) that is stochastic:

$$\Delta V_{\text{target}} = R_{\text{growth}} \text{binomial}[10, P_{\text{growth}}] \quad (4)$$

where the function $\text{binomial}[10, P_{\text{growth}}]$, implemented in the numpy package in Python, returns a random integer drawn from a binomial distribution for the number of successful growth steps (in the last 10) based on a single-step success probability P_{growth} . This stochastic function is used to limit the synchronicity of growth and subsequently triggered cell divisions. To keep cells well within the bounds of the simulation lattice, all growth ceases at 3000 MCS.

For mesenchymal cells, the base growth probability is regulated by juxtacrine signaling through EphrinB1–EphB2/3.³⁴ This is implemented by increasing P_{growth} from 10% to 50% for mesenchymal cells in contact with other mesenchymal cells. The growth rate is then regulated by BMP2, BMP4, and Noggin.⁴⁰ Noggin binds to BMP2 and BMP4 with a 1:1 stoichiometry and blocks their receptor binding sites.^{62,63} The model assumes that binding of Noggin to BMPs is very strong and saturated so that the amount of free BMP2 (i.e., that available to bind to its receptor) is simply computed as

$$[\text{BMP2}]_{\text{free}} = [\text{BMP2}]_{\text{max}} \left(0, \frac{[\text{BMP2}] + [\text{BMP4}] - [\text{Noggin}]}{[\text{BMP2}] + [\text{BMP4}]} \right) \quad (5)$$

and the free BMP2 concentration then determines the growth rate via a sigmoidal Hill function:

$$R_{\text{growth,MESENCH}} = (1.2 \text{ pixels/MCS}) \frac{[\text{BMP2}]_{\text{free}}}{[\text{BMP2}]_{\text{free}} + AC_{50,\text{BMP2}}} \quad (6)$$

Table 2. Cell Behaviors and Regulatory Signals Explicitly Represented in the Virtual Palate Mode

cell type	behavior	↑ or ↓ regulated By	references
mesenchymal	growth and proliferation	↑ contact with Mesench; BMP2 ↓ Noggin(i) ^a	30,34,40,57
all	apoptosis	↓ contact with any cell or ECM	76
all	ECM production	↑ contact with Mesench; BMP2 ↓ Noggin(i)	17,40
all	secretion of FGF10	↑ contact with Mesench; SHH	29,30,57
all	secretion of BMP2	↑ contact with Mesench; SHH	30,40,57
all	secretion of BMP4	↑ contact with Mesench; BMP4 ↓ SHH; Noggin(i)	40,57
all	secretion of MMPs		67
nasal only	secretion of FGF7	↑ FGF7 ↓ SHH	29
epithelial	growth and proliferation	↑ FGF10; FGF7; EGF ↓ TGFβ3	28,29,64
all	apoptosis	↑ TGFβ3 ↓ contact with any cell or ECM; contact w/medium or Peri; EGF	31,37,43–45,64,65
all	differentiation to Mesench	↑ contact w/Mesench or ECM; TGFβ3 ↓ contact w/medium or Peri; EGF	45
all	differentiation to periderm	↑ contact w/medium	77
all	BM production/maintenance	↑ contact w/Mesench or ECM ↓ MMPs	67,78
all	secretion of MMPs	↑ TGFβ3	33,67
all	secretion of SHH	↑ FGF10; BMP4 ↓ FGF7; Noggin(i)	29,30,36,39,40
medial only	secretion of TGFβ3	↓ EGF	41
medial only	secretion of EGF	↓ TGFβ3	42
oral only	secretion of Noggin		36
periderm	growth and proliferation	↑ FGF10; FGF7; EGF ↓ TGFβ3	28,29,64
all	apoptosis	↑ TGFβ3 ↓ contact with any cell or ECM; contact w/medium or Peri; EGF	31,37,43–45,64,65
all	polarization		77
all	motility	↑ TGFβ3 ↓ EGF	31
all	secretion of SHH	↑ FGF10; BMP4 ↓ FGF7; Noggin(i)	29,30,36,39,40
medial only	secretion of TGFβ3	↓ EGF	41
medial only	secretion of EGF	↓ TGFβ3	42

^aNoggin(i) refers to indirect regulation (i) by Noggin binding to BMP2 or BMP4 preventing activation of the BMP receptor as coded in the model.

R_{growth} and P_{growth} are used to calculate ΔV_{target} for each cell following eq 4 above. After updating a mesenchymal cell's target volume, its target surface area is updated accordingly.

For epithelial and periderm cells, the growth probability is always 50%. Their growth rate is upregulated by FGF10, FGF7, and EGF, and downregulated by TGFβ3.^{28,29,64} Both epithelial and periderm growth rates are determined by a similar combination of sigmoidal functions

$$R_{\text{growth,EPI/PERI}} = \max \left[0, R_0 + R_{\text{max,FGF}} \left(\frac{[\text{FGF10}]}{[\text{FGF10}] + \text{AC}_{50,\text{FGF10}}} + \frac{[\text{FGF7}]}{[\text{FGF7}] + \text{AC}_{50,\text{FGF7}}} \right) + R_{\text{max,EGF}} \left(f_{\text{change}} \frac{[\text{EGF}]^4}{[\text{EGF}]^4 + \text{AC}_{50,\text{EGF}}^4} - \frac{[\text{TGF}\beta 3]^4}{[\text{TGF}\beta 3]^4 + \text{AC}_{50,\text{TGF}\beta 3}^4} \right) \right] \quad (7)$$

with slightly different constants: $R_{\text{max,EGF}} = 0.08$ pixels/MCS for both, but $R_0 = 0.016$ and 0.01 pixels/MCS and $R_{\text{max,FGF}} = 0.2$ and 0.12 pixels/MCS for epithelial and periderm cells, respectively. After updating each

ectodermal cell's target volume according to eq 4, its other shape parameters are updated accordingly.

Cell division is allowed when a cell's volume exceeds its cell-type-specific mitosis trigger volume (V_{mitosis}). For most cell types, V_{mitosis} is set to 2× that cell type's initial average volume. The only exceptions are medial epithelial and periderm cells for which the multiplier is instead 3×, slowing proliferation along the medial edge. When mesenchymal cells reach their trigger volume, mitosis occurs immediately, but ectodermal cells will delay mitosis until sufficient time has passed since a particular cell's previous division. The wait time is initially set randomly to between 1 and 101 MCS (to randomize cell cycle phases) but is reset to 50 MCS after each cell division. Once a cell has committed to mitosis, all of its compartments divide, and its target volumes are conserved with a random 46.5 to 53.5% split between parent and daughter cell. The division plane is random for mesenchymal cells and along the minor axis of elongated epithelial and periderm cells. After division, shape parameters of the parent and daughter cell are updated according to cell type as described above. Although ECM and basement membrane elements are encoded in the model as generalized "cells", these do not correspond to biological cells and are thus not allowed to undergo cell division. Growth of ECM and the basement membrane is instead driven by secretion from MESENCH and EPI cells (see 2.e6–7).

2.5.2. Apoptosis. Cells in the model commit to apoptosis (programmed cell death) stochastically. The probability for any cell to make this commitment depends on juxtacrine signaling, encoded through the identity of that cell's neighbors, and on the local concentration of paracrine signals, in particular EGF and TGF β 3.^{31,37,43–45,64,65} Cellular commitment to apoptosis is irreversible. If any cell becomes isolated from all other cells, its apoptosis probability is 10% per MCS. In addition, the apoptosis probability for any epithelial cell or periderm compartment depends on the local concentration of EGF and TGF β 3 according to a pair of Hill functions:

$$P_{\text{apo,EPI/PERI}} = \max \left[0, P_{\text{max}} \left(\frac{[\text{TGF}\beta 3]}{[\text{TGF}\beta 3] + \text{AC}_{50,\text{TGF}\beta 3}} - f_{\text{change}} \frac{[\text{EGF}]}{[\text{EGF}] + \text{AC}_{50,\text{EGF}}} \right) \right] \quad (8)$$

where f_{change} is the fold-change in EGF receptor (EGFR) expression levels (e.g., induced by toxicant-driven AhR activation; see below), and P_{max} is the maximum apoptosis probability determined by the nature of the cell's microenvironment. For periderm cells, apoptosis probabilities are considered separately for their apical and basal compartments, but a decision point for apoptosis by either compartment commits the entire cell. For an apical periderm compartment, the default P_{max} is 0.05% per MCS and increases by 5% per MCS if it is not in contact with the medium and another 5% per MCS if it is not in contact with the basal periderm. P_{max} is determined similarly for a basal periderm compartment, but the increases occur if it is not in contact with the medium or apical periderm and if it is not in contact with the medium or any other periderm cell. For an epithelial cell, the default P_{max} is just 0.01% and increases by 1% per MCS if the cell is not in contact with the medium or periderm. Apoptosis decisions are evaluated for every cell every tenth MCS (with $P_{\text{apo}}(\Delta N = 10) = 1 - (1 - P_{\text{apo}})^{10}$).

As noted earlier, cells committed to the apoptotic pathway are not immediately removed but are flagged and have their target volume decreased every subsequent MCS by $\Delta V_{\text{target}} = -(1/\tau_{\text{apo}})V_{\text{target}}$. As a result, the target volume of dying cells decays exponentially with time constant $\tau_{\text{apo}} = 120$ MCS. Dying cells also round up due to a removal of their target length, if present, and by setting their target surface area to $4V_{\text{target}}^{1/2}$. Once V_{target} for a dying cell is below a few pixels, energy fluctuations can lead to its removal from the model. During palate fusion, when apoptosis is prevalent in the model, the two to three time constants required to completely eliminate an apoptotic cell correspond to approximately 2–6 h.

2.5.3. Cell Differentiation. After 200 MCS, epithelial cells can stochastically differentiate into either periderm or mesenchymal cells with probabilities that depend on the local microenvironment. Differentiation to the periderm can occur when an epithelial cell has lost contact with the underlying mesenchyme or its matrix. Such differentiation maintains a cell's oral, medial, or nasal designation. The probability for such an epithelial-to-periderm transition (EPT) *in silico* is 0.01% per MCS if the cell is already in contact with the periderm and increases to 1% per MCS if it is in direct contact with the surrounding medium. Similarly, dedifferentiation to the mesenchyme can occur when an epithelial cell has lost contact with the overlying periderm or surrounding medium but maintains contact with the mesenchyme or its matrix. This occurs most often as periderm cells die, and small "pearls" of epithelial cells remain in the region of the MES. The base epithelial-to-mesenchymal transition (EMT) probability is 0.01% per MCS and can increase by up to 0.1% depending on the local concentration of TGF β 3 and EGF as described by a pair of Hill functions:

$$P_{\text{EMT}} = \max \left[0, 0.01\% + 0.1\% \left(\frac{[\text{TGF}\beta 3]}{[\text{TGF}\beta 3] + \text{AC}_{50,\text{TGF}\beta 3}} - f_{\text{change}} \frac{[\text{EGF}]}{[\text{EGF}] + \text{AC}_{50,\text{EGF}}} \right) \right] \quad (9)$$

To track the prevalence of EMT, cells that undergo these transitions are designated as type MESENCH_EMT. For both types of differentiation, decisions are evaluated for every cell every 10th MCS (with $P_{\text{EPT or EMT}}(\Delta N = 10) = 1 - (1 - P_{\text{EPT or EMT}})^{10}$).

2.5.4. Periderm Polarization. The model includes periderm polarization by allowing each periderm cell to have a single matched pair of apical and basal compartments. These are treated as separate generalized "cells" in CompuCell3D but have the same cluster ID. The initial model configuration has no apical periderm, but any time after 50 MCS, unpaired basal compartments can nucleate apical compartments. Nucleation involves designating a location along the basal–periderm–medium border as the seed for a matching apical compartment and setting new target volumes: $V_{\text{target,A}} = 1.2$ (length of basal–periderm–medium border); and $V_{\text{target,B}}$ decreased by $V_{\text{target,A}}$. Target surface areas and lengths are then set accordingly to favor elongated shapes for both compartments. Unpaired apical compartments can similarly nucleate new basal compartments (anywhere along their border that is not in contact with surrounding medium), but this is a rare event that occurs only when a basal periderm compartment is stochastically lost during a simulation.

2.5.5. Cell Motility. In a cellular Potts models, a cell moves via stochastic fluctuations that add or remove lattice sites along its periphery. The fluctuation amplitude or effective temperature (T_{eff}) determines how likely fluctuations are when their pseudoenergy change is unfavorable (i.e., for $\Delta E > 0$, $P = e^{-\Delta E/T_{\text{eff}}}$) and thus determines how easy it is for a cell to move: cells with higher T_{eff} are thus more motile. In our model, T_{eff} is 4 for most cells and ECM elements but can vary for the periderm. After 300 MCS, the T_{eff} for periderm cells can range from 0 to 10 as determined by Hill functions of the local concentration of TGF β 3 and EGF:³¹

$$T_{\text{eff,PERI}} = \max \left[0, 2 + 8 \left(\frac{[\text{TGF}\beta 3]}{[\text{TGF}\beta 3] + \text{AC}_{50,\text{TGF}\beta 3}} - f_{\text{change}} \frac{[\text{EGF}]}{[\text{EGF}] + \text{AC}_{50,\text{EGF}}} \right) \right] \quad (10)$$

Periderm cells are thus fairly quiescent until TGF β 3 signals rise, and they then become highly motile.

2.5.6. ECM Secretion. Mesenchymal cells in the model also secrete ECM so that growth of the palatal shelves is a combination of cell proliferation and ECM accumulation.¹⁷ ECM secretion and ingestion are implemented as a transfer of target volume between a mesenchymal cell and an adjacent ECM element, if one exists; if not, a new ECM element is nucleated at a random location along the cell border. The volume transferred from each cell is based on its BMP2-dependent growth rate (from eq 6 above) and its ECM secretion probability ($P_{\text{sec ECM}}$):

$$\Delta V_{\text{to ECM}} = \min[V_{\text{cell}}, R_{\text{growth,MESENCH}}(-1 + 0.4 \text{binomial}[10, P_{\text{sec ECM}}])] \quad (11)$$

where the transfer volume cannot exceed the cell volume. The two terms, respectively, represent deterministic ECM ingestion and stochastic ECM secretion. The ECM secretion probability is regulated by juxtacrine signaling implemented by increasing $P_{\text{sec ECM}}$ from 25% to 50% for mesenchymal cells in contact with other mesenchymal cells. Ingestion and secretion are in balance (on average) for $P_{\text{sec ECM}} = 25\%$ in eq 11, but increasing this value to 50% clearly favors ECM secretion. After transferring $\Delta V_{\text{to ECM}}$, the shape parameters of each mesenchymal cell are updated based on its new target volume. As with cell growth, ECM secretion ceases at 3000 MCS to keep all cells and ECM well within the bounds of the simulation lattice.

2.5.7. BM Maintenance. The model includes an explicit basement membrane (BM) that underlies it and is maintained by epithelial cells. Individual BM elements are treated as generalized "cells" with a small target volume (3.5) and a large volume stiffness (40X that of epithelial cells). The large volume stiffness is needed to limit the rate at which BM elements are lost to stochastic fluctuations, and the half-integer target

volume is purposely chosen to allow these stiff elements to still move over the model's lattice, i.e., there is no energy penalty for changing a BM "cell's" volume from 3 to 4 or vice versa but a very large energy penalty for lowering the volume to 2 or less. To create an elongated BM chain (as seen in cross-section), BM elements are nucleated in pairs with matching cluster IDs, and each element is allowed to form one intracluster and one intercluster focal-point plasticity (FPP) link. FPP links are elastic connections between the centroids of BM elements that form stochastically (when allowed) and that break when stretched beyond a maximum length (see eq 3). The BM is maintained by a balance between nucleation of new BM elements, which occurs when an epithelial cell is in direct contact with the mesenchyme or its matrix, and BM degradation through the action of MMPs. This degradation occurs stochastically in proportion to the local concentration of MMPs ($P = 0.5\% [MMP]$) and is implemented by setting the target volume of a degraded BM element to zero. If a BM element becomes unpaired due to the loss of its partner from fluctuation or degradation, it is then also degraded by immediately setting its target volume to zero.

2.5.8. Signal Secretion. The final cell behavior is regulated secretion of diffusible signals. Since concentrations and secretion rates of these signaling molecules are normalized by their respective half-maximal activity, AC_{50} values in the above equations can generally be treated as equal to 1. The one exception arises from the direct binding of Noggin to BMP2 and BMP4,⁶² which requires identical normalization for all three concentrations. We normalize these to the minimum AC_{50} found in the model for any BMP2- or BMP4-regulated process (self-activation of BMP4 secretion). With this choice, the relevant AC_{50} is instead equal to 2 for BMP4-regulated secretion of SHH (Table 3) and for BMP2-regulated mesenchymal growth and ECM secretion (eq 6).

The secretion rate for a particular diffusible signal can be regulated by multiple other signals. Mathematically, the rate is determined by a sum of sigmoidal functions (Hill coefficient = 1), with one sigmoid for each regulating signal. Regulated secretion in the model is summarized in Table 3 with a listing of the coefficients of these sigmoidal functions.

2.6. Modeling Teratogenic Perturbations of the TGF β 3/EGF Switch. As noted in Figure 2 and Table 3, the actions of TGF β 3 and

EGF comprise a mutually inhibitory switch. To enable simulations of chemical exposures, we modeled this switch in greater detail as follows: (1) EGF binding to its receptor (EGFR) leads to a decreased rate of TGF β 3 expression; (2) TGF β 3 binding to its receptor (TGF β R) leads to a decrease of EGF expression; (3) these ligand–receptor bindings are fast processes compared to transcriptional rates; and (4) transcription–degradation processes of both pathways combine to yield a net cellular response. Ligand-binding of either growth factor [GF] to its cognate receptor $[R_{GF}]$ is thus described by

$$[GF :: R_{GF}] = [R_{GF}]_{\text{tot}} \frac{[GF]}{[GF] + K_d}$$

with transcription and degradation combining for each growth factor to yield

$$v_{GF} = v_m \frac{K_m^h}{K_m^h + [GF :: R_{GF}]^h} - k_d [GF]$$

The symmetry between EGF and TGF β 3 is maintained for most parameter values: dissociation constant $K_d = 0.71$; Michaelis constant $K_m = 1$; max transcription rate $v_m = 0.214$; Hill coefficient $h = 4$; and degradation rate $k_d = 0.07$. The symmetry was broken by either having different cognate receptor levels or different initial growth factor concentrations. The teratogenic effects simulated below were modeled by modifying EGF receptor levels.

3. RESULTS

3.1. Morphogenetic Outgrowth and Fusion. Simulations of growth and fusion in an idealized cross-section of the anterior palate ran for 6000 MCS, corresponding to mouse gestation days E12.5 to E16. Earlier prepattern events (E11.5 to E12.5) were imposed as initial conditions. Model outputs of the cell field for normal fusion are shown at various stages in Figure 3; model outputs for the signaling gradients are shown as snapshots at the midline fusion stage in Figure 4. The full time-dependent signaling gradients are shown alongside the cell field in Movie S1. As expected for a Monte Carlo simulation, the relationship between developmental stage and MCS is not linear, with the simulation slowing as development progresses. On the basis of the landmarks of first contact and the completion of fusion between E14.5 and E15.5, the simulation corresponds to gestational days E13, E14, E15, and E16 at approximately 25, 500, 2000, and 5000 MCS, respectively.

Prefusion outgrowth was driven primarily by internal expansion of the mesenchyme, initially via cell growth and proliferation but increasingly through ECM secretion at later times. Subepithelial accumulation of ECM in the model implicitly represents sulfated proteoglycans; hydration-induced expansion of the hyaluronate-rich matrix is not represented in the model but is left for future development.²⁵ Signaling between the mesenchyme and ectoderm allowed the epithelial and periderm layers to match mesenchymal growth, maintaining complete barrier coverage but not overproliferating.

Physical contact between right–left palatal processes occurred in the midline leading to the formation of a transitory MES (Figure 3). Across dozens of simulation runs, first contact and MES formation occurred at 1600–2000 MCS, corresponding in real time to E14.8 to E15.0. By 3000 MCS, breakdown of the MES was marked by a few remaining pearls of ectoderm. Such pearls are observed in cross-sectional images of mouse palates taken between E15–E15.5.¹⁸ By 5000 MCS (E16), mesenchymal confluence was normally complete.

Morphological changes in the MEE commenced with increased motility of periderm cells and programmed cell death. Both changes were semiautonomous, following increases

Table 3. Regulated Signal Secretion in the Virtual Palate Model^a

secretion of	by cells of	regulated by	with sigmoid coefficient (MCS ⁻¹)
SHH	epithelium and periderm	FGF10	5.0
		BMP4 (free)	2.5
		FGF7	−5.0
FGF10	mesenchyme	SHH	1.5 (0.2 if no contact with mesenchyme)
FGF7	nasal mesenchyme	FGF7	2.0
		SHH	−2.0
BMP2	mesenchyme	SHH	1.5 (0.2 if no contact with mesenchyme)
BMP4	mesenchyme	BMP4 (free)	4.5 (0.2 if no contact with mesenchyme)
		SHH	−2.0
Noggin	oral epithelium and periderm	basal	1
TGF β 3	medial epithelium and periderm	basal	0.214 + random[0.014]
		EGF	−0.214
EGF	medial epithelium and periderm	basal	0.214 + random[0.014]
		TGF β 3	−0.214
MMP	mesenchyme	basal	1
MMP	epithelium	basal	1
		TGF β 3	3

^aNegative sigmoid coefficients represent down-regulation with overall secretion rates limited to a minimum of zero. Note that the rate at which FGF10, BMP2, and BMP4 are secreted depends on whether the secreting mesenchymal cell is in contact with other mesenchymal cells.

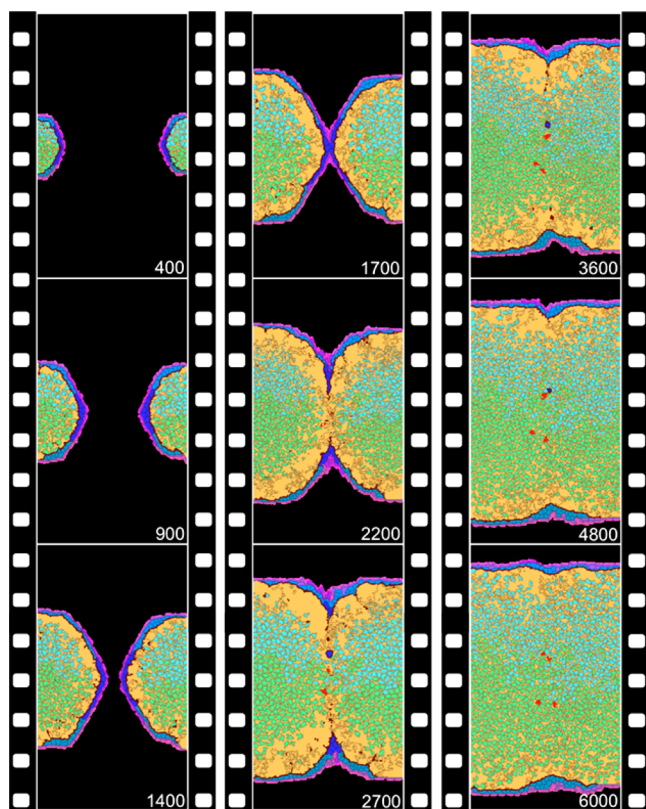


Figure 3. Outgrowth, midline contact, and fusion of the right–left palatal process simulated with the virtual palate model. Orientation and cell types are defined and color-coded as in Figure 1. Simulation time (in MCS) is noted at the bottom right of each frame and progresses from top to bottom along the three panels. Real time slows as the model progresses: E14–500 MCS, E15–2000 MCS, and E16–5000 MCS. Note MES formation and breakdown, leaving behind residual pearls (blue) and EMT-derived mesenchyme (red), as well as limited movement of interior mesenchymal cells. A complete set of time-lapse frames is available as [Movie S1](#).

in TGF β 3 levels, but not requiring physical contact between opposing palatal shelves.³³ Once contact did occur, the periderm and then epithelial cells underwent increased rates of cell death. This initial MES breakdown was followed by migration of much of the residual MEE to triangular regions near the oral and nasal surfaces. Such epithelial triangles are regularly observed *in vivo*.^{18,66} In the model, the migrating cells were not following a chemical gradient. Instead, migration to the epithelial triangles was an emergent phenomenon driven by differential cell–cell adhesion. By this point, the remnants of the MES were just a few pearls of epithelial cells. As the surrounding basement membrane was degraded by secreted MMPs, these MES pearls were eventually eliminated through a combination of cell death and EMT (red cells in Figure 3). With the probabilities chosen for each process in the model, approximately 70–85% of MEE cells were eventually cleared via cell death (many after initially migrating to the epithelial triangles). The remaining handful of cells were either incorporated into the oral/nasal ectoderm or underwent EMT to join the mesenchyme.

As the MES broke down and the mesenchyme became confluent, there was a slight mixing among mesenchymal cells from the right and left shelves. A similar degree of mixing is evident along the boundary of the oral and nasal mesenchyme, but there was little movement by the interior mesenchyme.

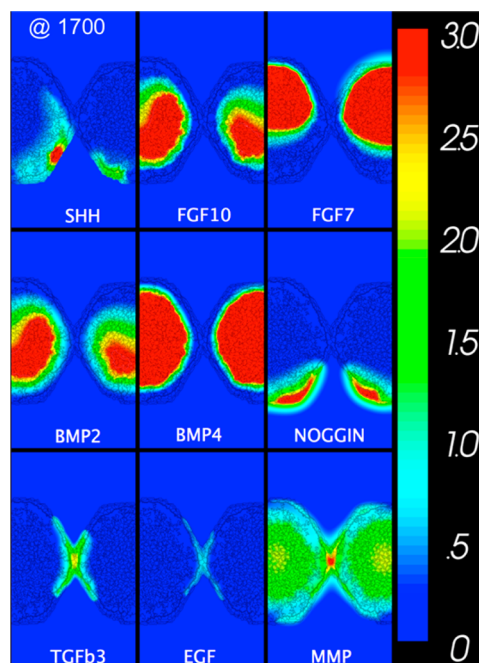


Figure 4. Biochemical signal gradients in the virtual palate model. All gradient fields are as labeled at the midline fusion stage (1700 MCS in Figure 3). The network driving the simulation is shown in Figure 2. The concentration of each effector molecule has been scaled relative to its minimum AC_{50} for biological effects (color bar to the right). A complete set of time-lapse gradients is available in [Movie S1](#).

Growth and proliferation in the mesenchyme also slowed as the MES broke down and fewer cells were available to secrete SHH. Despite this slowing, the few surviving MEE cells could secrete enough SHH to drive local mesenchymal growth. The model thus ceased all growth processes at 3000 MCS ($\sim$ E15.4), an artificial constraint imposed only to keep the final cell configuration from squeezing against the bounds of the modeled lattice. By the time this constraint took effect, fusion was essentially complete. The final period in the simulation (5000–6000 MCS) was not designed to recapitulate normal postfusion biology but was instead present to permit discrimination between inhibition versus delay of MES breakdown following various prefusion disturbances (described below).

3.2. Biochemical Gradients. Diffusion gradients of the model's explicit biochemical signals are shown as snapshots at the midline fusion stage in Figure 4 and dynamically for the entire simulation in [Movie S1](#). The patterns varied slightly between simulations and were similar but nonidentical when comparing the right and left processes in a single simulation. This highlights the model's inherent degree of stochasticity. The main driver of outgrowth was a positive feedback loop between SHH secreted from the MEE and FGF10 from the underlying mesenchyme. Differences in the spatial distribution of these signals' sources and in their relative diffusion kinetics created a broader paracrine domain for FGF10. As SHH expression decayed with dissolution of the MEE, so too did FGF10 expression. The model further captured the mutual inhibitory loop between SHH–FGF7 as well as the interaction between SHH and BMP signals. Note that mesenchymal proliferation and ECM production were influenced by the concentration of free BMP, i.e., that not bound to Noggin. Dissolution of the MES was regulated by the mutual inhibition of EGF and TGF β 3, signals coexpressed in the MEE. As shown at first contact, TGF β 3 signals were a few times

stronger than EGF signals. This disparity increased during midline fusion as the mutually inhibitory switch fell into a TGF β 3-high state. Matrix metalloproteinases (MMPs) are represented generically in the model. Some (e.g., MMP-13) are known to be highly expressed in the murine MEE and palatal mesenchyme in a manner dependent on TGF β 3 function.⁶⁷ In simulations, MMP was ubiquitously expressed at basal levels but was enhanced at sites of high TGF β 3 signaling. The spatial patterns of these key signals qualitatively matched those observed in coronal sections of mouse palates through a combination of immunostaining, *in situ* hybridization, and LacZ reporters: FGF10, FGF7;²⁹ SHH;³⁰ TGF β 3;^{33,68} Noggin;³⁶ SHH, FGF10, BMP2, and BMP4;⁴⁰ MMP;⁶⁷ EGF and TGF β .⁶⁹

3.3. Hacking the Control Network. To test functionality in the control network, simulations were run for *in silico* knockouts, i.e., by individually shutting off the secretion of each effector molecule after 200 MCS. This permitted the model to initiate normally and then simulate loss of function phenotypes. The final morphologies of the tested knockouts are shown in Figure 5.

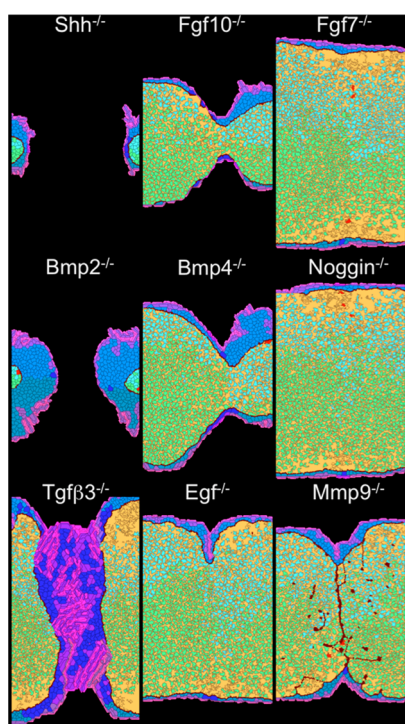


Figure 5. Growth and fusion defects for *in silico* knockouts. Each image shows model output at 6000 MCS (after E16) for loss of function of the indicated effector molecule. A time-lapse movie of the TGF β 3 knockout is available as Movie S2.

Loss of SHH or BMP2 function severely disrupted outgrowth and resulted in failure to achieve midline contact (Figure 5). Weaker phenotypes were seen in simulations with lost BMP4 or FGF10 function, which partially disrupted outgrowth but did not completely prevent contact or fusion. These phenotypes are consistent with an SHH-FGF10 loop acting through BMP2 and modulated by BMP4 as the major *in silico* drivers of outgrowth.^{29,30,40} In contrast, loss of function for FGF7 and Noggin had little to no impact on growth or fusion, yielding only slight differences in the relative proliferation of oral and nasal mesenchyme.

Loss of TGF β 3 function led to excessive MEE proliferation and a failure of tissue fusion. In contrast, loss of EGF function had little effect save a small midline indentation. These phenotypes are consistent with an EGF-TGF β 3 negative feedback loop being the key regulator of MES dissolution.^{41–44} These signals not only inhibited one another but also influenced MEE cell behaviors in a diametrically opposed manner: EGF signaling promoted MEE growth and survival, and TGF β 3 signaling promoted epithelial apoptosis, EMT, and motility (Figure 2). The two signals act like a switch (see below). Loss of EGF function only pushes the switch to its normal TGF β 3-high state, but loss of TGF β 3 function pushes the switch the other way and prevents the processes necessary for MES breakdown. Downstream of TGF β 3, loss of MMP function also prevents complete MES breakdown through its control of BM degradation.

3.4. Simulating the Teratogenic Response to Environmental Exposures. To investigate the predictive capacity of the virtual palate model to simulate adverse effects of environmental insult, we attempted to recapitulate the teratogenic action of 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD) in a chronic exposure scenario and of all-trans retinoic acid (ATRA) in an acute exposure scenario.⁵⁴ Among the components of the EGF and TGF β 3 pathways that regulate MES dissolution, TCDD or ATRA exposure leads to an overexpression of the EGF receptor (EGFR), which usually decreases in the MEE around E13.⁴⁶ We thus investigated the mutual inhibition of EGF and TGF β 3 in more detail using a model of EGF-TGF β 3 interactions described above in Methods and shown in Figure 6A. Overall, the EGF-TGF β 3 switch is modeled by bound TGF β 3:TGF β R suppressing EGF secretion and bound EGF:EGFR suppressing TGF β 3 secretion. Changes in receptor levels can be induced by exogenous agents to alter the switch's bias.

As then shown in the bifurcation plots in Figure 6B,C, the receptor expression levels determine the behavior of this biochemical circuit. For the base receptor levels used in all simulations above (1.86 and 2.14 for EGFR and TGF β R, respectively), the switch has a single steady state with EGF low and TGF β 3 high. This can be seen by looking at the steady state values for an EGFR fold-change of one in Figure 6B. If the EGFR level is then changed, the circuit has only TGF β 3-high steady states for fold-changes <1.1, only EGF-high (TGF β 3-low) steady states for fold-changes >1.3, and a bistable region in between. EGFR fold-changes in that bistable region could yield steady states with either TGF β 3 or EGF high. The one adopted by the system is history dependent. If the base receptor levels are increased (2.86 for both), the switch has a much wider bistable zone, spanning EGFR fold-changes from 0.8 to 1.8 as shown in Figure 6C. Given the central role of this biochemical switch in palate fusion, we investigate below how the symmetry between EGF-TGF β 3 pathways and the width of the switch's bistable zone impacts normal and toxicant-perturbed fusion in the palate model.

TCDD-induced cleft palate in the mouse is mediated by AhR with overproliferation of cells in the MEE leading to a failure of palate fusion.⁵⁴ To simulate the effect of chronic TCDD exposure scenarios on palate fusion, we parametrized TCDD-induced activation of AhR in terms of EGFR fold-changes, i.e., TCDD exposure was modeled synthetically as a direct increase in the expression of EGFR imposed by the degree of AhR activation (Figure 7A,B and Movies S3 and S4). Modeling the low-hysteresis switch described in Figure 6B (and used in all simulations above) showed a critical effect on MEE hyperplasia and MES breakdown as the EGFR fold-change increased from

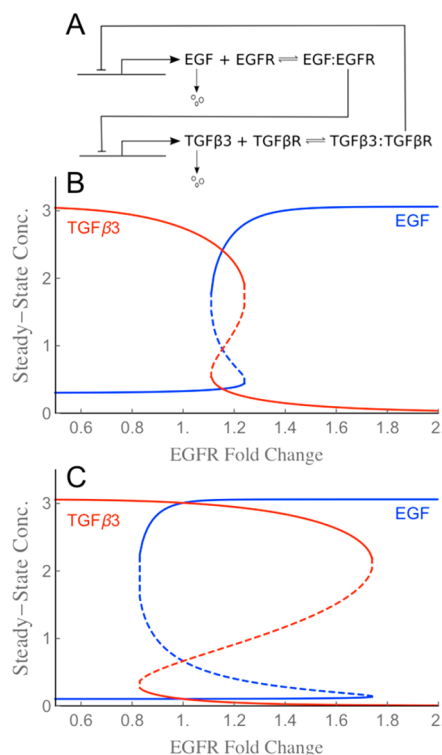


Figure 6. Bifurcation plots for the EGF-TGF β 3 circuit implemented in the virtual palate model. (A) Schematic of the mutual inhibitory circuit. Symmetry between EGF and TGF β 3, as well as the structure of the bistable zone, can be broken through parameter choices. (B) Steady-state behavior for base receptor levels that yield a narrowly bistable switch with low hysteresis. For any EGFR fold-change in the bistable zone from 1.1 to 1.3, the circuit has three steady state solutions (i.e., three solutions for EGF and three for TGF β 3): two of the solutions are stable (solid lines) and one unstable (dashed line). (C) Increasing and equating the base receptor levels yield a switch with a wider bistable zone (0.8 to 1.8) and higher hysteresis (unstable solutions dashed).

1.1 $\times$ to 1.2 $\times$ (Figure 7A). This result is consistent with the bifurcation plot computed for the isolated biochemical switch (Figure 6B). Modeling the high-hysteresis switch (described in Figure 6C) yielded similar consequences at slightly higher EGFR fold-changes (Figure 7B). Interestingly, the high-hysteresis switch also produced more cases with partial fusion (Table 4). This shows that bistability in the switch can be manifest as adjacent regions of the MES alternately adopting TGF β 3-high or EGF-high states.

In a second set of scenarios, we modeled acute exposure to ATRA. Similar to TCDD, ATRA induces an increase in EGFR.⁵⁴ Negative cross-talk between retinoid and TGF β 3 signaling is thus imposed synthetically in the model by an ATRA-mediated effect on EGF signaling that in turn suppresses TGF β 3 secretion (Figure 7C,D and Movies S5 and S6). The imposed time-dependent EGFR fold-changes are shown at the top of each panel, with expression levels increasing rapidly just after 1000 MCS and decaying back to normal with a time constant of 600 MCS. In simulations using the low-hysteresis switch (Figure 7C), an initial flip from TGF β 3-high to EGF-high occurred around 1100 MCS as the EGFR fold-change peaked (near 1.8 $\times$) and a switch back to TGF β 3-high occurred when the EGFR fold-change dropped back down below 1.1. Although fusion was delayed, it was eventually complete in 8 of 8 such simulations (Table 4). On the other hand, simulations using the high-hysteresis switch (Figure 7D) were unable to switch back to a

TGF β 3-high state (despite a lower maximum EGFR fold-change of just 1.6 $\times$) and failed to fuse in 4 of 4 cases. This is a hysteresis effect inherent in the bistability of the switch. Transient exposure simulations with the high-hysteresis switch were only able to complete fusion when the maximum EGFR fold-change was sufficiently low (1.2 or 1.3) that the circuit never entered an EGF-high state.

4. DISCUSSION

Computer simulations that predictively model morphogenetic fusion provide a resource to investigate the interplay of developmental pathways and processes, and assess the impact of environmental stressors on these processes. The virtual tissue model constructed here for the mouse secondary palate effectively simulated higher-level phenotypes (e.g., midline contact, MES breakdown, mesenchymal confluence, and fusion defects) in response to genetic or environmental perturbations. An *in silico* perturbation analysis of various control features revealed model functionality with respect to cell signaling systems and feedback loops for growth and fusion, diverse individual cell behaviors, and collective cellular behaviors leading to physical contact and midline fusion, and quantitative analysis of the TGF/EGF switch that controls MES breakdown, a key event in morphogenetic fusion. This computer model adds to similar systems models toward a growing “virtual embryo” toolbox for simulation and quantitative prediction of adverse developmental outcomes following genetic perturbation and/or environmental disruption.^{21,22} Simulating a complex embryological system in this manner has potential applications for high-throughput hypothesis-based testing and translating chemical-biological interactions into tissue-level predictions.

Several strengths of the virtual palate model include the ability to forward-engineer biological circuits underlying growth and fusion defects of various types following inactivation of specific elements of the signaling network. This included quantitative analysis of cell signaling systems for growth and fusion, including positive (SHH-FGF10) and negative (EGF-TGF β 3) feedback loops mediating epithelial-mesenchymal interactions. *In silico* knockouts or “cyber-morphs” provide an indication of the specificity of the underlying signaling systems as they are currently understood. SHH-FGF signaling, for example, is an important mediator of epithelial-mesenchymal interactions during outgrowth of the palatal processes,¹⁹ and this circuit was represented in the molecular network. On the other hand, a novel SHH-FOXF-FGF18 circuit in the palate development molecular network was recently demonstrated.⁷⁰ That circuit regulates FOXF-dependent palatal mesenchymal growth downstream of SHH signaling, partly by repressing mesenchymal expression of *Fgf18*, which is an inhibitor of *Shh* expression. This newer information is not yet implemented in the virtual palate model. The model can, however, be continually refined to incorporate new molecular information and knowledge, making it a “living document” that gets updated as our understanding of the biology improves.

Individual cellular behaviors such as differential adhesion, cell motility, proliferation, EMT, and apoptosis coded into the model led to collective cellular behaviors that enable physical contact and midline fusion as higher order (emergent) properties. These emergent behaviors can then be traced back to the underlying molecular control circuits, for example, the TGF/EGF switch that controls MES degeneration.⁵⁴ The computational model simulated the critical balance in relative levels of TGF β 3 and EGF signaling in the MEE and MES breakdown. Flipping the

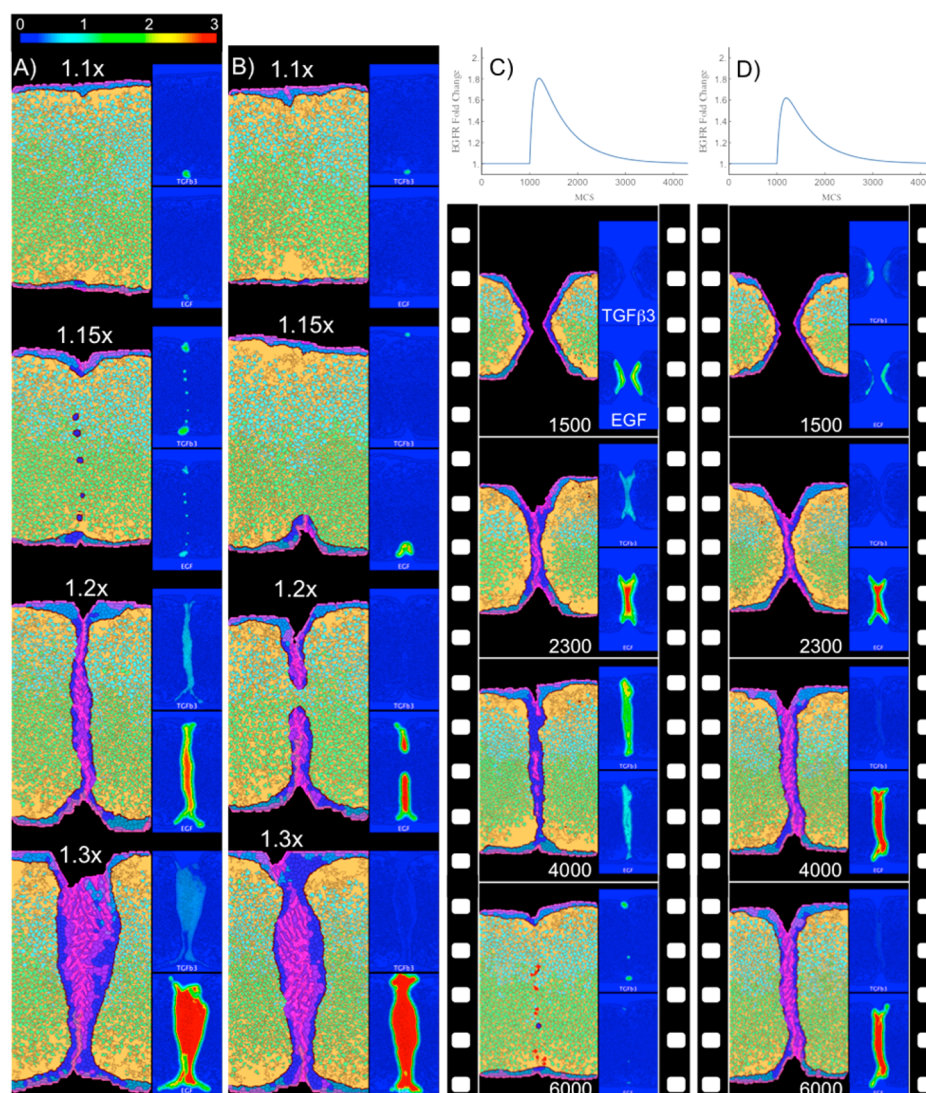


Figure 7. Simulations of toxicant-induced fusion defects in the virtual palate model. (A,B) Final states for simulations of chronic TCDD exposure. Scenarios were parametrized via AhR-mediated fold-changes in EGFR (as indicated from 1.1 $\times$ to 1.3 $\times$) for (A) the low-hysteresis version of the EGF-TGF β 3 switch ($n = 54$, tipping point $\sim 1.2\times$) and (B) the high-hysteresis version ($n = 34$, tipping point $\sim 1.2\times$). Insets show corresponding EGF and TGF β 3 signal gradients. The phenotype of a thickened MES preventing mesenchymal confluence is similar in appearance to the histology of palatal shelves from a TGF β 3 knockout (with or without additional Alk mutations).⁴⁴ (C,D) Time-lapse images from simulations of transient acute exposure to ATRA parametrized as time-dependent EGFR fold-changes (top). Palate fusion was delayed using the low-hysteresis switch ($n = 24$, tipping point $>1.8\times$) (C), but failed using the high-hysteresis switch ($n = 16$, tipping point 1.5 $\times$) (D) despite being subjected to a smaller maximum EGFR fold-change. Complete time-lapse image sets are available for specific examples as [Movies S3](#) (panel A, 1.15 $\times$), [S4](#) (panel B, 1.2 $\times$), [S5](#) (panel C), and [S6](#) (panel D).

TGF β 3/EGF switch on or off in the MEE provided a singular heuristic approximation for an MES-breakdown phenotype. This speaks to the unique advantage of cell agent-based modeling for systems toxicology in predicting the tissue-level impact of a localized molecular lesion. Furthermore, as demonstrated by the panel of “cybermorphs” derived from more generally hacking the control network, *in silico* manipulation of the biological network exposes the system to different “what-if” scenarios that can be cell type-specific and/or stage-dependent, turning genes up or down and in different combinations to probe the sensitivity of the system to potential molecular initiating events that could disrupt palatal growth and fusion.

Signaling through the EGFR is a key event in at least some forms of teratogen-induced cleft palate.⁵⁴ Functional EGFR receptors are expressed in the mouse palatal epithelium as early as E12 and sensitize the MEE to hyperplasia instead of

breakdown.⁷¹ Chemicals that increase EGFR expression, induce MEE proliferation, and disrupt fusion include retinoic acid, hydrocortisone, and TCDD.^{28,46,69,72} For example, litters dosed with TCDD (24 $\mu\text{g/kg}$) or ATRA on E12 had fetuses where the palatal processes came in contact with one another, but MEE cells continued to express high levels of EGFR, and the palatal processes failed to fuse.⁴⁶ New insights into the EGF-TGF β 3 switch governing the clearance of the MES can be gauged from the linkage of the cell agent-based simulations with a formal hysteresis switch for transient exposures to teratogens. This could guide the design of experimental studies to measure and re-engineer the molecular and cellular parameters governing the complexity of this switch in the context of systems toxicology and synthetic biology.⁷³ Multiple runs of both the low- and high-hysteresis models predicted a tipping point in the system controlling MES breakdown when chronic TCDD exposure

Table 4. Palatal Fusion Outcomes from Simulated Exposures

exposure model	hysteresis switch	max EGFR fold-change	# fused cases <i>n</i> (%-fused)	# partially fused <i>n</i> (%-fused)	# unfused cases <i>n</i> (%-fused)
constant	low	1.10×	15 (100%)	0	0
		1.15×	15 (100%)	0	0
		1.20×	0	0	21 (100%)
		≥1.30×	0	0	3 (100%)
constant	high	1.0×	15 (100%)	0	0
		1.2×	2 (17%)	9 (75%)	1 (8%)
		1.3×	0	0	5 (100%)
		1.4×	0	0	15 (100%)
transient	low	1.2×	6 (100%)	0	0
		1.3×	8 (100%)	0	0
		1.6×	2 (100%)	0	0
		1.8×	8 (100%)	0	0
transient	high	1.2×	4 (100%) ^a	0	0
		1.3×	3 (100%) ^a	0	0
		1.6×	0	1 (25%)	3 (75%)
		1.8×	0	0	4 (100%)

^aMinor indentations noted.

induced as little as a 1.2-fold increase in EGFR expression. In contrast, the impact of acute ATRA exposure (through EGF-mediated functional inhibition of TGFβ3 signaling) had a higher tipping point. The greater tolerance to transiently increased EGF signaling was especially evident in the low-hysteresis switch model where its reversibility pushed the tipping point above a 1.8-fold increase in EGFR.

The capacity of the computational model to quantitatively output both low-level (e.g., cell fate and behavior) and high-level (e.g., growth dynamics and emergent behaviors) features of the system provides a direct linkage to address how much change in different pathways would elevate risk for cleft palate. The virtual palate model identified the critical roles of a number of the effector molecules as cybermorph phenotypes that recapitulated *in vivo* models. It also identified critical points of imbalance in the control network driving palatal development that serve as potential molecular initiating events for adverse outcome pathways (AOPs) leading to cleft palate. Nonetheless, there may be additional AOPs leading to cleft palate that are not captured by the model, either because the present two-dimensional model does not address some known aspects of the physical biology, e.g., reorientation, postfusion osteogenic differentiation, or regional differences in mesenchymal cell orientation between oral–nasal sides or anterior–posterior axes,^{74,75} or because alterations of the cellular/molecular mechanisms leading to palate fusion may be compensated by unknown interactions that prevent the manifestation of a morphological defect. As such, future improvement in the virtual palate model should strive toward a more realistic representation of the ECM and mechanical forces in 3D and carefully compare how perturbations alter fusion in the model versus *in vivo*. Quantifying those differences will be the key to identifying key compensatory mechanisms.

Understanding how tissues and organs are shaped during embryogenesis is a central question in developmental biology and a challenge for predicting developmental toxicity. The virtual palate model adds to similar systems-based models that ultimately lead to a “virtual embryo” for computer simulation and quantitative prediction of adverse developmental outcomes. Although the current computational model was constructed

from published information in the mouse, it can be used as a heuristic approximation to translate *in vitro* data generated from human cell-based studies such as ToxCast. This helps integrate the very large body of research at the cellular–molecular scale, and the equally large body of knowledge at the organ–embryo scale both clinically and more recently through physics-based investigation utilizing novel bioengineering, biomechanics, and microphysiological systems.

■ ASSOCIATED CONTENT

§ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acs.chemrestox.6b00350](https://doi.org/10.1021/acs.chemrestox.6b00350).

Legends to the movie files (PDF)

Cellular dynamics for simulations covering normal palate fusion (AVI)

Disrupted fusion caused by TGFβ3 knockout (AVI)

Effects of exposure to subcritical chronic exposure to TCDD (AVI)

Effects of exposure to supracritical chronic exposure to TCDD (AVI)

Effects of exposure to subcritical transient exposure to ATRA (AVI)

Effects of exposure to supracritical transient exposure to ATRA (AVI)

Archive of the complete python code for the model (ZIP)

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Notes

The views expressed in this article are those of the authors and do not necessarily reflect the views or policies of the U.S. Environmental Protection Agency. Mention of trade names or commercial products does not constitute endorsement or recommendation for use.

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■ ABBREVIATIONS

2D, two-dimensional; 3D, three-dimensional; AC_{50} , half-maximal activity concentration; AhR, aryl hydrocarbon receptor; AOP, adverse outcome pathway; ATRA, all-trans retinoic acid; BM, basement membrane; BMP, bone morphogenetic protein; CC3D, CompuCell3D software; ΔE , change in pseudoenergy function; δ , Kronecker delta function; ΔN , duration in MCS; E11.5 to E16.0, embryonic days 11.5 to 16.0 in mouse; ECM, extracellular matrix; EGF, epidermal growth factor; EGFR, EGF receptor; $E_{interface}$, E_{shape} , and $E_{focal-point\ plasticity}$, terms in pseudoenergy function; EMT, epithelial-to-mesenchymal transition; EphB, ephrin receptor B; EPITH, epithelial cell type; EPT, epithelial-to-periderm transition; EXT, external fluid; f_{change} , fold-change; FGF, fibroblast growth factor; FGFR2b, fibroblast growth factor receptor 2b; FOXF, forkhead box protein F; FPP, focal-point plasticity; J_{int} and J_{ext} , matrices of cell-type-dependent interfacial energies; K_d , ligand–receptor binding affinity; L , cell's major axis length; λ_{link} , stiffness parameter for focal-point plasticity links; λ_{vol} , $\lambda_{surface}$, and λ_{length} , stiffness parameters of shape energy terms; MCS, Monte Carlo steps; MEE, medial edge epithelium; MES, medial edge seam; MESENCH, mesenchymal cell type; MMP, matrix metalloproteinase; P , probability; PERI_A, PERI_B, periderm cell type - apical or basal compartment; Ptc1, patched (receptor for SHH); r , centroid-to-centroid distance; R_{growth} , growth rate; R_0 and R_{max} , basal and maximal growth rates; ρ , generalized cell ID; σ , cluster ID; S , cell surface area; SHH, sonic hedgehog; Smo, smoothened (downstream effector of SHH signaling); τ_{apo} , time constant for apoptotic cell death; τ , cell type index; TCDD, 2,3,7,8-tetrachlorodibenzo-p-dioxin; T_{eff} , fluctuation amplitude or effective temperature; TGF β , transforming growth factor beta; TGF β R, transforming growth factor beta receptor; V , cell volume

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